

C Program - Day-4 Class Assignment

1. Write a program to read a string from the user and display its length using `strlen()`.
2. Write a program to input a string and copy it into another string using `strcpy()`. Display both the original and copied strings.
3. Write a program to input two strings and compare them using `strcmp()`. Display whether they are equal or not.
4. Write a program to input two strings and join them using `strcat()`. Display the resulting string.
5. Write a program to input two strings. Compare their lengths using `strlen()` and display the longer string.
6. Assume the correct password is "admin123". Read a password from the user and compare it using `strcmp()`. Display "Access Granted" or "Access Denied".
7. Write a program to input a filename and append the extension .txt using `strcat()`.
8. Write a program to input two strings, concatenate them using `strcat()`, and display: the concatenated string and the total number of characters using `strlen()`
9. Write a C program that fulfills the following requirements:
Input: The program should accept a string of characters.
Vowel Array: Define a character array containing both lowercase and uppercase vowels ('a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U').
Processing: Iterate through each character of the input string and compare it against the characters in the vowel array.
Output: Print each character from the input string that is a vowel, followed by a space.
10. WAP a C program that accepts a string, then prints each character one by one while also identifying whether it's a lowercase letter, an uppercase letter, or a digit or a special character.
11. Problem Statement: Simple C Login System
Develop a C program that simulates a basic user login system. The program should perform the following tasks:
Register a User: Prompt the user to set a username and a password. Store these credentials in character arrays.
Login: After registration, prompt the user to log in by entering their username and password.
Validate Credentials: Compare the entered login credentials with the stored registration credentials.
Display Outcome: Print a success message if the credentials match or a failure message if they do not.
12. Problem Statement: Secure Password Creator

Create a C program that acts as a secure password creator. The program must repeatedly prompt the user to enter a new password until it meets all of the following security criteria:

Length: The password must be at least 8 characters long.

Uppercase: It must contain at least one uppercase letter (A-Z).

Lowercase: It must contain at least one lowercase letter (a-z).

Digit: It must contain at least one digit (0-9).

Special Character: It must contain at least one special character from the set !@#\$%^&*()_+=[{}|;:'",.<>/?.

For each attempt that fails to meet the criteria, the program should provide specific feedback to the user, listing exactly which requirements were not met. Upon successful creation, the program should confirm that the password has been set.

13. Secure User Authentication

Create a C program that implements a simple user authentication system. The system should first guide a user through a **registration process** where they must create a username and a password that meets specific security criteria.

The password criteria are as follows:

Minimum **8 characters** in length.

Contains at least one **uppercase letter**.

Contains at least one **lowercase letter**.

Contains at least one **digit**.

Contains at least one **special character** from the set !@#\$%^&*()_+=[{}|;:'",.<>./?.

The program should use a loop to repeatedly prompt the user for a password until all criteria are met. Once the password is valid, the program should proceed to a **login process**, where it prompts the user to enter the previously set username and password.

Finally, the program must validate the entered login credentials against the stored registration data and display a message indicating whether the login was successful or failed.

Note: The solution should use string-handling functions from the <string.h> and <ctype.h> libraries to perform the necessary checks.

MCQ questions

1. What is the output of the following code?

```
char str[] = "Computer";  
printf("%d", strlen(str));
```

- A) 7
- B) 8
- C) 9
- D) 10

Answer:

2. What is the output of the following code?

```
char s1[20] = "Good";  
char s2[] = "Day";  
strcat(s1, s2);  
printf("%s", s1);
```

- A) Good
- B) Day
- C) GoodDay
- D) DayGood

Answer:

3. What is the output of the following code?

```
printf("%d", strcmp("Apple", "Apple"));
```

- A) 1
- B) -1
- C) 0
- D) Undefined

Answer

4. What is the output of the following code?

```
char str[] = "Hello";  
printf("%c", str[1]);
```

- A) H
- B) e
- C) l
- D) o

Answer:

5. Which of the following correctly declares a string in C?

- A) `char str = "Hello";`
- B) `string str = "Hello";`
- C) `char str[] = "Hello";`
- D) `str char[] = "Hello";`

Answer:

6. What is the output of the following code?

```
char s1[20] = "Hi";  
char s2[] = "There";  
strcpy(s1, s2);  
printf("%s", s1);
```

- A) Hi
- B) There
- C) HiThere
- D) Error

Answer:

7. What is the output of the following program?

```
char fruits[][10] = {"Apple", "Mango", "Orange"};  
printf("%s", fruits[1]);
```

- A) Apple
- B) Mango
- C) Orange
- D) Error

Answer:

8. What is the output of the following program?

```
char city[][10] = {"Delhi", "Mumbai", "Chennai"};  
printf("%c", city[2][3]);
```

- A) n
- B) e
- C) n
- D) a

Answer:

9. What is the output of the following program?

```
char name[2][20] = {"Alice", "Bob"};  
strcpy(name[1], "John");  
printf("%s", name[1]);
```

- A) Alice
- B) Bob
- C) John
- D) AliceJohn

Answer:

10. What is the output of the following program?

```
char str[3][20] = {"C", "Programming", "Language"};
strcpy(str[0], str[1]);
strcat(str[0], str[2]);
printf("%s\n", str[0]);
printf("%s\n", str[1]);
```

- A) ProgrammingLanguage
Programming
- B) CLanguage
Programming
- C) Programming
Language
- D) ProgrammingLanguage
Language

Answer:

Explain

1. **Why is the null character ('\0') important in a string? Explain with an example.**
2. **Differentiate between the following string functions with suitable examples and how it works:**
 strlen()
 strcpy()
 strcmp()
 strcat()
3. **Explain the difference between comparing two strings using the == operator and using the strcmp() function.**
4. **What is a two-dimensional character array (array of strings)? Explain its declaration and initialization with an example.**
5. **Explain How Strings are Stored in a Two-Dimensional Character Array**

```
char cities[3][10] = {"Delhi", "Mumbai", "Chennai"};
```

What happens if a string longer than 9 characters is stored in one of the rows?